



Comsats University Islamabad, Lahore Campus

Terminal Examination – FALL 2020

Course Title	Information Security	Course Code	CSC432	Credit Hours	3(2,1)
Instructor/s	Dr. Tariq Umer, Dr. Muhammad Sharjeel	Program Name	BSSE		
Semester	8 th	Batch	FA16	Section	A,B
Time Allowed	3 hours	Maximum Marks			
Student Name:		Registration No.			

Important Instructions/Guidelines:

1. Avoid writing long stories.
2. Use of S-Box, Lookup table cheat sheet is allowed
3. Exchange of calculator is not allowed
4. Match of solution from a certain threshold will result in failure of entire course!
5. Return the question paper along with the answer sheet. **Missing paper will result in elimination of 5 marks**

Question no. 1

[3+5+3+3+3+3]; PLO-1; CLO :<2,3>; Bloom Taxonomy Level: <Analyzing>

1. You need to guess a password having the following characteristics:

- It may possibly consist of all the lower-case letters (A-Z)
- It may possibly consist of all the upper-case letters (a-z)
- It may possibly consist of all the digits (0-9).
- The length of the password is 7 characters.

How many combination you need to try through brute force in the worst case.

2. Generate the first round key of DES using the following P-box, if the initial key is **FF-FF-FF-FF-FF-FF-FF-FF** in hexa-decimal
3. Applying Affine cipher, decrypt the following text when $a = 7$, $b = 5$
ICH OPIPUH JB RHI JS RZPU GZDHU
4. 8839 and 8849 are two prime numbers. Their product is 78216311. Find all the factors of 78216311.
5. Why TGS is required while AS is already authenticating the client in Kerberos?
6. Alice sends a public key to Bob digitally signed by CA₁, while Bob's public key is digitally signed by CA₂. Bob's system rejected the Alice public key. Why? What is the solution to this problem?

Question no. 2

[3+4+4+2+2]; PLO-4; CLO :<3,4>; Bloom Taxonomy Level: <Applying>

1. What are the two conditions for any encryption scheme to be computationally secure?
2. Apply a scenario of Diffie-Hellman protocol with $g=6$ and $p=13$. Assume, Alice picks 7 as her private number while Bob picks 8 as his private number. Now assume that Trudy launches a man in the middle attack and picks 6 as her private number. As you can guess that in this case, Alice and Bob will communicate through Trudy so please identify the value of shared secrets between Alice and Trudy as well as Bob and Trudy following the Diffie-Hellman message exchange.
3. Describe the key development mechanism using RSA when the values of p and q are given to be 7 and 11 respectively? Although there are multiple options for choosing the encryption value 'e', let's choose 7 that fulfill the criteria and then find 'd' accordingly. You should provide all details and outcome should be written in the form of private and public keys. Demonstrate the working of your system by encrypting a number and then retrieving it back using decryption.
4. User A wants to download an operating system image from the vendor website. On downloading the OS image, the user should be able to assess the integrity and authenticity of the OS image in an optimal way. Draw a diagram to illustrate the process to ensure and validate integrity and authenticity of the OS image. Briefly describe each step of the process illustrated in your diagram.

5. In a database of user accounts, following row exist: (ID=21, digest=ab4jp, salt=2287). A user tries to login with username 426 and password p3ach. How will the system checks if user's credentials are correct. Does this scenario is vulnerable to dictionary attack? Reason?

Question no. 3

[2+2+2+4]; PLO-3; CLO :<1,4>; Bloom Taxonomy Level: <Remembering>

- A) You have designed a new algorithm for checking message authenticity by message digest. According to the algorithm, we can add together the numeric values of the alphabetical index of the first letter of each word, where letter "A" has index 0 and "Z" has index 25. As an example, "I graduated from COMSATS" will produce MD 21(as I=8, G=6, F=5, C=2 and $8+6+5+2=21$).
- You then use this algorithm to generate the digest of the following message: "I insist staying in COMSATS". Your friend launched a birthday attack and found a second candidate message having same digest. Can you identify/investigate the second message? Show your calculations.
- B) You then change the algorithm by replacing ADD with MULTIPLY. However, now the number of candidate messages increase from the previous case. Can you identify/debate the reason, why?
- C) You then change the algorithm described in part a) by adding the first and second letter from each word (For example, "Graduated from COMSATS" will produce 61, as G=6, R=17, F=5, R=17, C=2 and O=14 ($6+17+5+17+2+14 = 61$)). Can you analyze the effect of this modification on the strength of the algorithm by concluding whether the number of candidate messages will increases or decreases from the case in part a), why?
- D) Explain how the public and private key of the sender and receiver will be used for the following:

	Confidentiality	Integrity	Authentication	Non-repudiation
Sender +				
Sender -				
Receiver +				
Receiver -				

Question no. 4

[2+3]; PLO-4; CLO :<4>; Bloom Taxonomy Level: <Understanding>

- A) Hassan, a final-year university student, finds a way to access his university's grading system using a script he found online. Without authorization, he enters the system and checks the grade records of multiple students, including those of his competitors. He does not change any data but takes screenshots and shares them in a private WhatsApp group to mock others. One of the students reports the incident to the university IT department. After investigation, they trace the access back to Hassan's IP address.

Based on this scenario, which cyber law(s) under PECA has Hassan violated? Explain briefly.

- B) A company's IT helpdesk received a call from someone claiming to be a senior manager who forgot their password. The caller was very convincing and even knew the manager's employee ID and department. The helpdesk agent, wanting to help quickly, reset the account and shared the temporary password over email. Later that day, sensitive project files were accessed and downloaded from the manager's account without their knowledge. Upon investigation, it was found that the real manager had never requested a password reset.
1. What type of security vulnerability is this?
 2. Suggest one or two effective solutions to prevent this type of attack in the future.